Pseudo-Label Guided Bidirectional Discriminative Deep Multi-View Subspace Clustering

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***Abstract*—In practical applications, multi-view subspace clustering is hindered by data noise that disrupts the ideal block-diagonal structure of self-representation matrices, thereby degrading performance. Moreover, many existing methods rely solely on sample features, overlooking the valuable structural information in affinity matrices (e.g., pairwise relationships). While conventional contrastive learning strategies often introduce false negative pairs due to noise and unreliable sample selection. To address these challenges, we propose a pseudo-label guided bidirectional discriminative deep multi-view subspace clustering method (PBDMSC). Our approach first employs pseudo-label guided contrastive learning, using previous cluster assignments to select reliable positive and negative samples, which mitigates incorrect pairings and enhances low-dimensional representations. Then, a discriminative self-representation learning method is introduced that leverages pseudo-labels to enforce homogeneous expression constraints and incorporates a bidirectional attention mechanism to preserve the structured information from affinity matrices, thereby enhancing robustness. Experimental results on six real-world datasets demonstrate that our proposed method achieves state-of-the-art clustering performance.**

***Index Terms*—Multi-view clustering, pseudo-graph, contrastive learning, auto-encoder, self-expression.**

I. Introduction

MULTI-VIEW data is common in everyday life, where an object can be described through images, text, attributes, etc [1], [2], [3]. Each view captures distinct characteristics [4], [5], while there is also consistency across views, reflecting the same object. The key to multi-view clustering lies in how to extract consistent information from multiple views to achieve better clustering results [4], [6], [7]. For example, CvLP-DCL [8] proposes a framework that decomposes the label space into consensus and view-specific components, preserving both shared semantics and view-discriminative patterns. Similarly, MAFN [9] introduces cross-view attention to fuse spatial transcriptomics data adaptively, integrating intra- and inter-view features. In subspace clustering, latent cluster structures are identified using self-representation coefficients that encode linear relationships among samples [10]. Multi-view subspace clustering methods achieve cross-view feature fusion by learning independent self-representation matrices for each view [4], [11] or constructing shared matrices [12].

However, multi-view data is often contaminated by noise, which disrupts the block diagonal property (BDP) of the self-representation matrix and adversely affects subsequent clustering results [10], [13]. Noise-induced perturbations cause interference among self-representation coefficients across different subspaces, leading to nonzero entries in off-diagonal blocks. Inspired by the robust representation theory of affine subspaces [13], we observe that when sample distributions deviate from the ideal linear subspace assumption, conventional constraint methods fail to adequately capture the local structure within each subspace. To address this issue, we propose a pseudo-label guided homogeneity constraint strategy. Specifically, we employ a pseudo-label matrix $\mathbf{P}$ as a dynamic subspace indicator and explicitly cut off cross-subspace representation paths using the formulation $\mathbf{C} = \mathbf{C} \odot \mathbf{P}$. This strategy helps maintain an approximate block-diagonal structure in the self-representation matrix even in noisy environments.

From the perspectives of multi-source information fusion and graph signal denoising theory [14], [15], most existing multi-view subspace clustering methods mainly focus on learning the self-representation matrix from sample features [11], [12], [16]. They often overlook the rich structural information encapsulated in the affinity matrix [17], [18], [19]. The affinity matrix is commonly used in spectral clustering methods to construct consistent matrix for fusing multi-view information. Generally, it encodes the global pairwise similarities among samples and captures the global manifold structure more effectively. In contrast, the self-representation matrix is better suited for describing local linear relationships. We contend that integrating these two types of information can more comprehensively reveal the underlying structure of the data and further enhance the noise robustness of self-representation learning.

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The code is available at https://github.com/usualheart/PBDMSC.

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Moreover, multi-view data is typically high-dimensional and exhibits non-linear characteristics. To better address these challenges, neural networks have been widely applied to multi-view clustering tasks [20], [21], [22]. Typical approaches first employ autoencoders to project high-dimensional data into a low-dimensional space [16], [23], followed by clustering algorithms. Recent studies have further incorporated contrastive learning to improve the discriminative power of the learned representations [24], [25]. However, conventional contrastive learning methods often rely on k-nearest neighbors (kNN) to select positive and negative sample pairs [16], a strategy that may introduce false negatives due to local similarity estimation errors caused by noise. While DealMVC [26] attempts to mitigate this issue via pseudo-label guidance, it primarily focuses on sample features, which somewhat limits the quality of pseudo-labels and subsequent contrastive comparisons.

To address these challenges, we propose a pseudo-label guided bidirectional discriminative deep multi-view subspace clustering method (PBDMSC). By establishing a link between contrastive learning and subspace clustering through pseudo-label, our approach jointly enhances low-dimensional representation learning, self-representation learning, and clustering performance. Overall, our main contributions are as follows:

- We design a pseudo-label guided multi-view contrastive learning mechanism. By selecting positive and negative samples based on the pseudo-label graph, we can better mitigate the negative impact of incorrect pairs and learn more discriminative low-dimensional representations.
- We propose a discriminative self-representation learning method. By constraining the self-representation process to homogeneous samples using pseudo-labels and incorporating a bidirectional attention mechanism to capture both feature and structural consistency, we jointly promote more robust self-representation learning.
- Extensive experimental results on multiple benchmark datasets demonstrate that our proposed method outperforms current state-of-the-art multi-view clustering approaches.

The rest of this paper is structured as follows. Section II provides the related work. Section III introduces the proposed method, comprising three key components: multi-view encoder, pseudo-Label guided contrastive learning, and discriminative self-representation learning. Section IV presents the experimental evaluation and results. Finally, Section V concludes the paper.

II. Related Work

In this section, we first introduce subspace clustering and multi-view subspace clustering, followed by a discussion of recent advancements in learning discriminative self-representation matrices.

A. Subspace Clustering

Most subspace clustering models can be unified into a framework consisting of two steps: first, learning the self-representation matrix $\mathbf{C}$; then, performing spectral clustering to obtain clustering results. The learning process can be formalized as:

$$\underset{\mathbf{C}\in\mathbb{R}^{n\times n}}{\text{minimize}} \quad \frac{1}{2}\|\mathbf{X}-\mathbf{X}\mathbf{C}\|_F^2+\lambda g(\mathbf{C}) \tag{1a}$$

$$\text{s.t.} \quad \text{diag}(\mathbf{C})=\mathbf{0} \tag{1b}$$

where $\mathbf{X}\in\mathbb{R}^{d\times n}$ is the original data matrix, with each column representing a sample $\mathbf{x}_i\in\mathbb{R}^d$. $\mathbf{C}\in\mathbb{R}^{n\times n}$ is the self-representation matrix, whose $(i,j)^{th}$ element indicates the contribution of the i^{th} sample in reconstructing $\mathbf{x}_j$. $g:\mathbb{R}^{n\times n}\rightarrow\mathbb{R}$ denotes the regularization function. $\lambda>0$ is a fixed parameter to adjust the influence of the regularization term. The constraint in (1b) is an additional condition to avoid obtaining the trivial solution $\dot{\mathbf{C}}=\mathbf{I}_n$.

The purpose of the regularization term is to impose certain properties on $\mathbf{C}$, such as sparsity [10], [27], [28] and low-rankness [29], [30], [31], [32]. In recent years, methods based on neural networks have gradually been applied to the subspace clustering problem. [33] first proposed a Deep Subspace Clustering model (DSC), which uses a neural network-driven autoencoder to learn a low-dimensional embedding of the data, and then learns the self-representation matrix $\mathbf{C}$ on this low-dimensional embedding. [21] implicitly imposed a rank constraint on the self-representation matrix $\mathbf{C}$ based on DSC, learning low-rank representations (LRR) of the input data.

B. Multi-View Subspace Clustering

Since the input data for the multi-view subspace clustering problem comes from multiple views, the key challenge lies in learning consistent self-representation relationship. [4] believed that the self-representation matrices $\mathbf{C}$ from different views have similar block structures, and proposed to use a unified indicator matrix $\mathbf{F}$ to learn a consistent clustering result across multiple views. [34] proposed to integrate multiple affinity graphs into a consistent one based on topological consistency. [7] believed that existing multi-view clustering methods only consider learning a consistent clustering structure from multiple spectral embeddings or multiple affinity matrices, and proposed a bidirectional attentive method that simultaneously considers spectral embeddings and affinity matrices to explore a consistent subspace. Similarly, [35] learned a bidirectionally fused consistent embedding by simultaneously focusing on spectral embeddings and projection embeddings. In terms of neural network-based multi-view subspace clustering, [36], [37] extended the autoencoder to multiple views based on the DSC model, and combined contrastive learning to learn more discriminative multi-view low-dimensional embeddings. DealMVC [26] facilitates the learning of better representations via pseudo-label guidance.

C. Learning Discriminative Self-Representation Matrices

Research on learning discriminative self-representation matrices has explored various strategies, including enforcing sparsity, low-rankness, and other structural properties. The sparsity of the self-representation matrix helps enhance the robustness of the clustering model, for example, [28] imposed an $\ell_{2,1}$-norm to learn a sparse projection matrix, mitigating the negative impact of low-discriminative views. The reason for learning low-rank

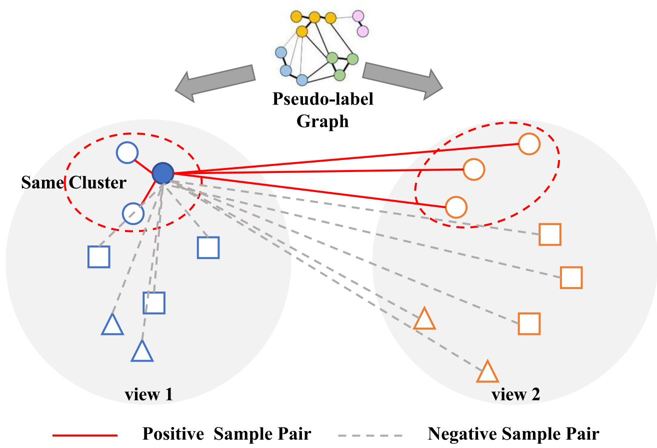


Fig. 1. Pseudo-Label Guided Contrastive Learning. The pseudo-label graph $\mathbf{P}$ (constructed from cluster assignments) encodes global class membership, guiding reliable positive/negative pair selection to mitigate false negatives. P_{ij} indicates whether samples i and j belong to the same class. Contrastive learning includes both within-view and cross-view comparisons, and through these two modes, it can better learn the consistent representations across multiple views.

self-representation matrices is that data is usually assumed to come from several low-rank subspaces [29], [32]. Many methods have been proposed for low-rankness, such as LRR [29], [30], [31] which solves for the low-rank representation $\mathbf{C}$ by imposing a nuclear norm constraint on $\mathbf{C}$. Furthermore, [13] argued that existing subspace learning methods overlook the issue of interference between samples from different affine subspaces. To address this, they proposed constraining the self-representation to occur only among homogeneous samples. [38] proposed to use pseudo-labels to guide the LDA model and learn a more discriminative low-dimensional subspace.

However, existing methods predominantly emphasize shallow linear subspace models, with limited application of deep neural networks. When it comes to employing deep neural networks for multi-view subspace clustering, there remain significant shortcomings in effectively learning more discriminative self-representation matrices. To address this gap, we propose a novel approach that leverages pseudo-labels in conjunction with neural networks to learn a more discriminative multi-view consistent self-representation matrix. Our method simultaneously prioritizes the extraction of low-dimensional embeddings and the construction of affinity matrices across multiple views.

III. Proposed Method

In this section, we formally introduce our proposed PBDMSC model. The model consists of a multi-view encoder module, pseudo-label guided contrastive Learning module and discriminative self-expression Learning module. The overall architecture of the model is shown in Fig. 2. Table I presents the main notations used throughout the paper.

TABLE I
Mainly Used Notations

Notation	Definition
$V, c, n,$	The number of views, clusters, and samples
d_v	The dimension of the v-th view
d	The dimension of low-dimensional representation
$\mathbf{x}_i^{(v)}$	The feature vector of the i-th data point in the v-th view
$\mathbf{C} \in \mathbb{R}^{n \times n}$	Consensus self-representation matrix
$\mathbf{A} \in \mathbb{R}^{n \times n}$	Consensus affinity matrix generated by $\mathbf{A} = \frac{\lvert\mathbf{C}\rvert + \lvert\mathbf{C}^T\rvert}{2}$
$\mathbf{S}^{(v)}$	The affinity matrix of the v-th view
$\mathbf{X}^{(v)} \in \mathbb{R}^{d_v \times n}$	The data matrix of the v-th view
$\mathbf{Z}^{(v)} \in \mathbb{R}^{d \times n}$	The low-dimensional representation of the v-th view
$\widehat{\mathbf{X}}^{(v)} \in \mathbb{R}^{d_v \times n}$	The reconstructed data matrix of the v-th view
$\mathbf{F} \in \{0,1\}^{n \times c}$	The clustering results
$\mathbf{P} \in \{0,1\}^{n \times n}$	The pseudo-label graph

A. Multi-View AutoEncoders

We adopt multiple autoencoders to learn the low-dimensional representation of data, where each encoder transforms the original multi-view data $\mathbf{X}^{(v)} \in \mathbb{R}^{d_v \times n}$ into a low-dimensional representation $\mathbf{Z}^{(v)} \in \mathbb{R}^{d \times n}$. The decoder reconstructs the low-dimensional representation $\mathbf{Z}^{(v)} \in \mathbb{R}^{d \times n}$ into multi-view data $\widehat{\mathbf{X}}^{(v)} \in \mathbb{R}^{d_v \times n}$. The autoencoder reconstruction loss is as follows:

$$\mathcal{L}_{AE} = \min_{\theta,\sigma} \sum_{v=1}^{V} \frac{1}{nd^{(v)}} \left\| \mathbf{X}^{(v)} - \widehat{\mathbf{X}}^{(v)} \right\|_F^2 \tag{2}$$

B. Pseudo-Label Guided Contrastive Learning

Traditional contrastive learning methods applied to deep clustering usually use k-nearest neighbors to select positive and negative samples, which relies on the computation of sample similarity graphs. At the same time, since k-nearest neighbors only consider the neighborhood information of a single sample, lacking the overall structural information between samples, it is easy to have false negative samples. Moreover, k introduces an additional hyperparameter, and different k's have a greater impact on the results. To solve the above problems and further learn more discriminative low-dimensional representations, we propose a pseudo-label driven contrastive learning module. The pseudo-labels come from the clustering results $\mathbf{F} \in \{0,1\}^{n \times c}$ generated in the previous training epoch, based on which we construct the pseudo-label matrix $\mathbf{P} \in \{0,1\}^{n \times n}$:

$$\mathbf{P} = \mathbf{F}\mathbf{F}^T \tag{3}$$

P_{ij} indicates whether samples i and j belong to the same class. The pseudo-label matrix $\mathbf{P}$ is updated every few training epochs based on the latest clustering assignments to progressively enhance its quality. As a graph constructed based on the clustering results from the previous round, $\mathbf{P}$ reflects the overall class division information of the samples. Therefore, selecting positive and negative samples based on the pseudo-label graph $\mathbf{P}$ can better avoid the issue of false negatives. Additionally, compared to the k-nearest neighbors method, this approach avoids introducing extra hyperparameters while achieving better performance.

Pseudo-label guided contrastive learning is illustrated in Fig. 1. The selection of positive and negative samples includes two modes: within-view and cross-view. Combining pseudo-label graph $\mathbf{P}$ and multi-view relationships, the method for determining positive and negative samples is as follows:

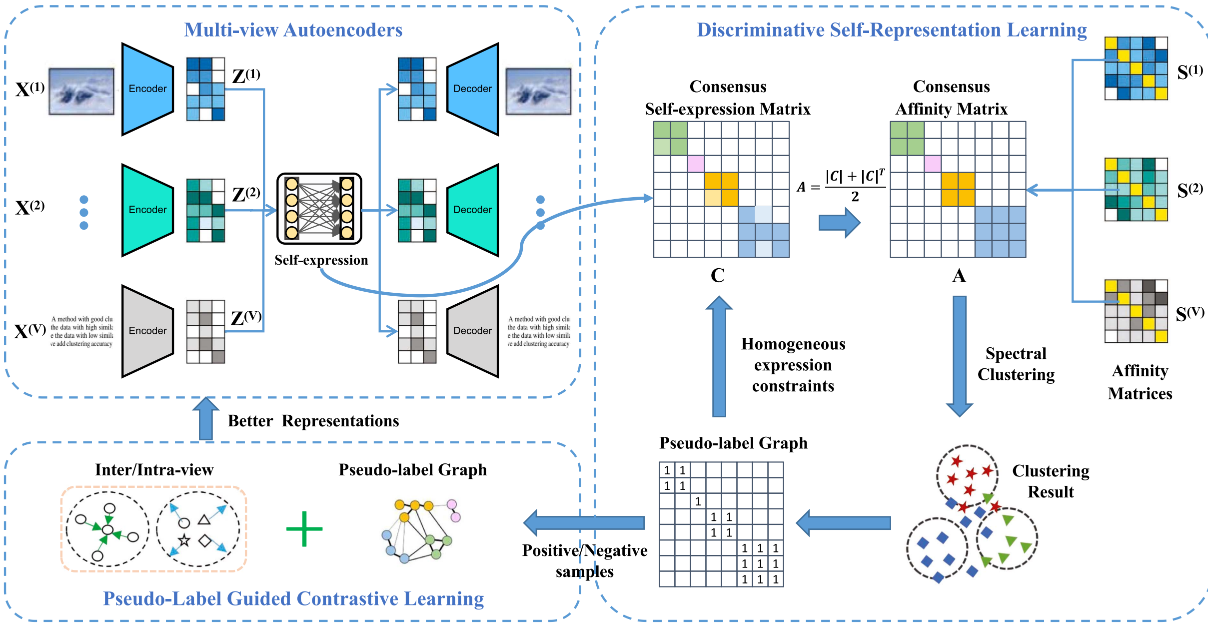


Fig. 2. The overall architecture of PBDMSC. Multi-view data $\mathbf{X}^{(v)}$ is transformed into low-dimensional representations $\mathbf{Z}^{(v)}$ through view-independent autoencoders. By applying discriminative self-representation learning on $\mathbf{Z}^{(v)}$, the model learns a self-representation matrix $\mathbf{C}$ that simultaneously emphasizes the affinity matrices from each view while restricting self-representation to homogeneous samples, thereby yielding a more pronounced block-diagonal structure. Finally, by incorporating a pseudo-label driven contrastive learning mechanism, the model refines both $\mathbf{Z}^{(v)}$ and $\mathbf{C}$ into more discriminative representations, ultimately enhancing clustering performance.

- *Positive Samples*: Representations of the same sample across different views, as well as those from samples sharing the same class label, are considered positive pairs.
- *Negative Samples*: Representations from samples belonging to different classes are treated as negative pairs, regardless of whether they originate from the same or different views.

The similarity between samples is calculated as follows:

$$sim(\mathbf{z}_i, \mathbf{z}_j) = e^{\frac{\left|\mathbf{z}_i^T \mathbf{z}_j\right|}{\|\mathbf{z}_i\|\|\mathbf{z}_j\|}} \tag{4}$$

where $\mathbf{z}_i$ is the low-dimensional representation of sample i under the autoencoder. For each sample $\mathbf{z}_i$, we define the following normalized similarity measures:

Intra-view positive similarity:

$$\overline{sim}_i^{intra\_positive} = \frac{1}{V\,|\mathcal{P}_i|} \sum_{v=1}^{V} \sum_{j \in \mathcal{P}_i} sim\left(\mathbf{z}_i^{(v)}, \mathbf{z}_j^{(v)}\right), \tag{5}$$

where $\mathcal{P}_i = \{j \mid P_{ij} = 1\}$ denotes the set of indices of samples sharing the same pseudo-label as sample i.

Inter-view positive similarity:

$$\overline{sim}_i^{inter\_positive} = \frac{1}{V(V-1)} \sum_{v=1}^{V} \sum_{u \neq v} sim\left(\mathbf{z}_i^{(v)}, \mathbf{z}_i^{(u)}\right). \tag{6}$$

Negative similarity:

$$\overline{sim}_i^{negtive} = \frac{1}{V^2\,|\mathcal{N}_i|} \sum_{v=1}^{V} \sum_{u=1}^{V} \sum_{j \in \mathcal{N}_i} sim\left(\mathbf{z}_i^{(v)}, \mathbf{z}_j^{(u)}\right), \tag{7}$$

where $\mathcal{N}_i = \{j \mid P_{ij} = 0\}$ denotes the set of indices of samples with a different pseudo-label from sample i.

The final contrastive loss function integrates these terms to jointly maximize positive similarities and suppress negative ones:

$$\mathcal{L}_{PCL} = -\frac{1}{n} \sum_{i=1}^{n} \log \frac{\overline{sim}_i^{intra\_positive} + \eta\, \overline{sim}_i^{inter\_positive}}{\overline{sim}_i^{negtive}}, \tag{8}$$

where η balances the contributions of intra-view and inter-view positive similarities, and n is the total number of samples. Through iterative clustering to obtain pseudo-labels and contrastive learning, these processes reinforce one another. As the pseudo-labels improve, the reliability of both positive and negative samples increases, thereby reducing the impact of false negatives.

C. Discriminative Self-Representation Learning

We adopt a unified self-representation matrix $\mathbf{C}$ to learn the consistent self-expression property across multiple views. First, we define the basic self-expression objective as follows:

$$\begin{aligned} &\min_{\mathbf{C}} \sum_{v=1}^{V} \alpha^{(v)} \left\|\mathbf{Z}^{(v)} - \mathbf{Z}^{(v)}\mathbf{C}\right\|_F^2 + \|\mathbf{C}^T\|_{2,1} \\ &\text{s.t.} \sum_{v=1}^{V} \alpha^{(v)} = 1,\ \text{diag}(\mathbf{C}) = 0 \end{aligned} \tag{9}$$

where $\alpha^{(v)}$ is the weight parameter for the loss of each view, designed to balance differences across views. It is learned through backpropagation as part of the neural network. The constraint

$\mathrm{diag}(\mathbf{C}) = 0$ is optionally used to eliminate a trivial solution of $\mathbf{C} = I$. To enhance robustness to noise, we impose a sparsity constraint on the columns of $\mathbf{C}$ via the $l_{2,1}$-norm, resulting in the term $\|\mathbf{C}^T\|_{2,1}$. This helps to represent using as few samples as possible, which contributes to improving robustness to noise.

1) Bidirectional Attention to Preserve Multi-View Structural Information: The self-representation learning model shown in (9) learns the self-representation matrix $\mathbf{C}$ solely based on the sample low-dimensional representation $\mathbf{Z}^{(v)}$, while ignoring the attention to each view's sample affinity matrix, which typically represents the structural information of the view. Since the matrix $\mathbf{C}$ can be transformed into the affinity matrix $\mathbf{A} = \frac{|\mathbf{C}|+|\mathbf{C}^T|}{2}$, to further preserve the structural information from each view, we first construct each view's affinity matrix $\mathbf{S}^{(v)} = \mathbf{D}^{(v)-\frac{1}{2}}\mathbf{K}^{(v)}\mathbf{D}^{(v)-\frac{1}{2}}$, where $\mathbf{K}^{(v)}$ is computed with entries $\mathbf{K}_{ij}^{(v)} = \exp(-\|\mathbf{x}_i^{(v)} - \mathbf{x}_j^{(v)}\|^2/(2\sigma^{(v)^2}))$. Here, $\sigma^{(v)}$ is empirically set as the average pairwise distance $\frac{1}{n}\sum_{k=1}^{n}\|\mathbf{x}_i^{(v)} - \mathbf{x}_k^{(v)}\|$ and $\mathbf{D}^{(v)}$ is the degree matrix of $\mathbf{K}^{(v)}$. We then constrain $\mathbf{A}$ to approximate $\mathbf{S}^{(v)}$ in the F-norm, ensuring that $\mathbf{A}$ inherits a consistent multi-view affinity structure to preserve cross-view structural information, thereby promoting $\mathbf{C}$ to exhibit the same property. At this point, the joint objective can be written as:

$$\begin{aligned}&\min_{\mathbf{C}} \sum_{v=1}^{V} \alpha^{(v)} \left\|\mathbf{Z}^{(v)} - \mathbf{Z}^{(v)}\mathbf{C}\right\|_F^2 + \gamma \sum_{v=1}^{V} \beta^{(v)} \|\mathbf{A} - \mathbf{S}^{(v)}\|_F^2 \\ &\quad + \|\mathbf{C}^T\|_{2,1} \quad \text{s.t.}\, \mathrm{diag}(\mathbf{C}) = 0, \\ &\quad \sum_{v=1}^{V} \alpha^{(v)} = 1, \sum_{v=1}^{V} \beta^{(v)} = 1, \mathbf{A} = \frac{|\mathbf{C}| + |\mathbf{C}^T|}{2}, \end{aligned} \tag{10}$$

Here, $\beta^{(v)}$ is the weight parameter for the consistent affinity matrix loss of each view, which is similarly learned through backpropagation.

To address this issue, we propose a pseudo-label guided homogeneity constraint strategy. Specifically, we employ a pseudo-label matrix $\mathbf{P}$ as a dynamic subspace indicator and explicitly cut off cross-subspace representation paths using the formulation $\mathbf{C} = \mathbf{C} \odot \mathbf{P}$. This strategy helps maintain an approximate block-diagonal structure in the self-representation matrix even in noisy environments.

2) Homogeneous Expression Constraints: Samples of different classes can be viewed as belonging to different subspaces. Self-representation across different subspaces typically constitutes noise interference. From a geometric perspective, the sparsest representation c_j should include only data points from the same subspace as x_j; that is, $c_{ij} \neq 0$ if and only if x_i and x_j belong to the same subspace. In this ideal scenario, the matrix $\mathbf{C}$ exhibits a clear block-diagonal structure, implying perfect clustering performance. To approach this as closely as possible, we propose a pseudo-label guided homogeneity constraint strategy. Specifically, we propose to constrain the matrix $\mathbf{C}$ using the pseudo-label graph $\mathbf{P} \in \{0,1\}^{n\times n}$ (from (3)).

$$\mathbf{C}_{ij} = 0 \text{ for } \mathbf{P}_{ij} = 0 \tag{11}$$

$\mathbf{P}_{ij} = 0$ indicates that samples i and j do not belong to the same class. Setting $\mathbf{C}_{ij} = 0$ ensures that self-representation only occurs between samples in the same subspace, and samples in different subspaces are as independent as possible. This helps to avoid interference from heterogeneous samples.

To seamlessly integrate this constraint into the objective function, we redefine the self-representation matrix as $\mathbf{C} \odot \mathbf{P}$ (where $\odot$ denotes element-wise multiplication). By combining the aforementioned terms, we derive the final loss function for discriminative self-representation learning:

$$\begin{aligned}\mathcal{L}_{DSL} = &\min_{\mathbf{C}} \sum_{v=1}^{V} \alpha^{(v)} \left\|\mathbf{Z}^{(v)} - \mathbf{Z}^{(v)}\left(\mathbf{C} \odot \mathbf{P}\right)\right\|_F^2 \\ &+ \gamma \sum_{v=1}^{V} \beta^{(v)} \|\mathbf{A} - \mathbf{S}^{(v)}\|_F^2 \\ &+ \left\|\left(\mathbf{C} \odot \mathbf{P}\right)^T\right\|_{2,1}, \text{s.t.} \quad \mathrm{diag}\left(\mathbf{C}\right) = 0, \\ &\sum_{v=1}^{V} \alpha^{(v)} = 1, \quad \sum_{v=1}^{V} \beta^{(v)} = 1, \\ \mathbf{A} = &\frac{\left|\mathbf{C} \odot \mathbf{P}\right| + \left|\left(\mathbf{C} \odot \mathbf{P}\right)^T\right|}{2}. \end{aligned} \tag{12}$$

D. Unified Formula

The final objective loss function is as follows:

$$\mathcal{L} = \min_{\theta.\sigma.\mathbf{C}} \mathcal{L}_{AE} + \lambda_1 \mathcal{L}_{PCL} + \lambda_2 \mathcal{L}_{DSL} \tag{13}$$

which includes the autoencoder reconstruction loss $\mathcal{L}_{AE}$, contrastive learning loss $\mathcal{L}_{PCL}$ and self-representation learning loss $\mathcal{L}_{DSL}$. Parameters λ_1 and λ_2 are used to balance the impacts of the three components. $\theta = \{\theta^v\}_{v=1}^{V}$and $\sigma = \{\sigma^v\}_{v=1}^{V}$ represent the parameters of the multimodal encoder and decoder, respectively. $\mathbf{C} \in \mathbb{R}^{n\times n}$ is the parameter of the consistent self-representation matrix. All parameters are trained through backpropagation.

IV. EXPERIMENTS

In this section, we evaluate the effectiveness of the proposed PBDMSC method.

A. Dataset

We conducted experiments on six real-world datasets, including COIL20 [39], MSRC-v6 [40], 3Sources[1], Digit4k[2], Caltech101-7 [41], and Caltech101-all [41]. Here are the detailed information of the datasets:

- COIL20: This is a widely used object image dataset with 20 classes. Each class has 72 image samples, resulting in a total of 1440 samples.
- MSRC-v6: This is an object image dataset with 7 classes, namely airplane, cow, tree, building, face, car, and bicycle.

[1]http://mlg.ucd.ie/datasets/3sources.html
[2]http://yann.lecun.com/exdb/mnist/

TABLE II
STATISTICAL INFORMATION OF THE DATASETS

Dataset	Samples	Clusters	Views	Features
COIL20	1,440	20	3	1024, 3304, 6750
MSRC-v6	210	7	6	1302, 48, 512, 100, 256, 210
3Sources	169	6	3	3560, 3631, 3068
Digit4k	4,000	4	2	30, 30
Caltech101-7	1,474	7	6	48, 40, 254, 1984, 512, 928
Caltech101-all	9,144	102	6	48, 40, 254, 1984, 512, 928

The dataset contains a total of 210 samples, with 30 samples per class.

- 3Sources: This is a text dataset consisting of 169 news articles from BBC, Reuters, and Guardian, with 6 classes representing different news topics.
- Digit4k: This is a digit dataset composed of 4000 digits, with 4 classes.
- Caltech101-all: This is a dataset specifically designed for object detection tasks, containing a total of 9144 image samples from 102 object classes.
- Caltech101-7: This is a subset of the Caltech101-all dataset, containing 7 object classes. The dataset has a total of 1474 image samples.

The statistical information of the datasets is summarized in Table II.

B. Comparison Methods

We compared our PBDMSC with nine different methods, including two subspace-based multi-view clustering algorithms, MLRSSC [11] and FPMVS_CAG [12], to evaluate the effectiveness of the proposed multi-view subspace learning. Additionally, we included two classical multi-view clustering algorithms, GMC [42] and FDAGF [43]. Furthermore, we utilized four contrastive learning-based multi-view deep clustering algorithms, MCGC [24], MFLVC [25], DealMVC [26] and GCCL [16], to assess the effectiveness of the proposed pseudo-label guided contrastive learning module. Lastly, we employed the affinity matrix-based multi-view deep clustering algorithm DMVSC [22] and the bidirectional probabilistic subspace multi-view clustering method BPSA [35] to evaluate the effectiveness of the proposed bidirectional attention module and discriminative self-representation learning module.

C. Experimental Setup

In our experiments, each view utilized separate encoder and decoder networks, which consisted of linear layers and ReLU activations. The decoder structure is the inverse of the encoder structure. We adopted a two-phase training approach: pretraining of the autoencoder network was conducted according to (2), while the second phase based on (12) and (8). The experiments use the Adam optimizer with a learning rate of 0.002 and a maximum training epoch of 300. The experimental environment consisted of a server equipped with an Intel Xeon Gold 6240 CPU and 4 NVIDIA GeForce RTX 3090Ti GPUs. The overall training process of the PBDMSC is shown in Algorithm 1.

Algorithm 1: PBDMSC.

Input: Input V views dataset $\mathbf{X}^{(1)}, \mathbf{X}^{(2)}, \ldots, \mathbf{X}^{(V)}$ and the number of clusters, c;
Calculate the affinity matrix $\mathbf{S}^{(v)}$ for each view based on $\mathbf{X}^{(v)}$;
Pre-train the multi-view autoencoder according to Eq. (2);
Randomly initialize the self-expression layer parameter $\mathbf{C}$, the view weight parameters $\alpha^{(v)}$ and $\beta^{(v)}$;
while *not converged or reached the maximum training epochs* **do**
  Calculate the overall loss according to Eq. (13) and train the entire PBDMSC network;
  Every certain number of epochs, use $\mathbf{C}$ to compute the clustering result $\mathbf{F}$ as pseudo-labels through Algorithm 2, and update the pseudo-label graph $\mathbf{P}$ according to Eq. (3);
Output: Output the clustering result $\mathbf{F}$;

Algorithm 2: Compute Clustering Results.

Input: Self-representation matrix $\mathbf{C}$, pseudo-label matrix $\mathbf{P}$, number of clusters c, intrinsic dimension of subspaces q
$\mathbf{A} = \frac{1}{2}(|\mathbf{C} \odot \mathbf{P}| + |\mathbf{C} \odot \mathbf{P}|^\top)$;
Compute truncated SVD: $\mathbf{A} = U_{1:m}\Sigma_{1:m}V_{1:m}^\top$ where $m = c \cdot (q+1)$;
Let $Q = U_{1:m}\Sigma_{1:m}^{1/2}$;
Build spectral affinity matrix: $A_{\text{clustering}} = (QQ^\top)^\alpha$;
Output: Clustering result $\mathbf{F}$ obtained by spectral clustering on $A_{\text{clustering}}$

After obtaining the self-representation matrix $\mathbf{C}$, we apply a classical post-processing method (Algorithm 2) to enhance its block structure and compute clustering result. Here, α is empirically selected to control the noise level, and q is the maximum intrinsic dimension of the subspaces [44]. For these parameters, we use the same settings as the classical subspace clustering algorithm DSC [33].

Hyperparameters were determined using a grid search strategy. The sensitivity of the model to hyperparameters will be discussed in the parameter analysis Section. For baseline methods, we used the default parameter settings from each paper as experimental parameter configurations and recorded the best results from different parameter combinations. To alleviate the

TABLE III
AVERAGE CLUSTERING RESULTS ON ALL DATASETS

Datasets	MLRSSC	GMC	MCGC	MFLVC	FPMVS_CAG	FDAGF	DMVSC	DealMVC	GCCL	BPSA	PBDMSC
ACC											
COIL20	0.7542	0.7910	0.7764	0.7514	0.5486	0.7597	0.8458	0.5979	0.8236	0.8549	**1.0000**
MSRC-v6	0.8571	0.8952	0.8286	0.8238	0.8619	0.9190	0.8000	0.8429	0.9190	0.9143	**0.9571**
3Sources	0.6213	0.6923	0.6331	0.6982	0.3254	0.7278	0.7929	0.7633	0.7930	0.7041	**0.9408**
Digit4k	0.7145	0.6600	0.5488	0.7655	0.8255	0.6465	0.6578	0.8665	0.7785	0.9173	**0.9230**
Caltech101-7	0.6174	0.6920	0.6750	0.8073	0.5495	0.7035	0.8450	0.8236	0.7408	0.6947	**0.8711**
Caltech101-all	0.1365	0.1950	0.2501	0.1834	0.2591	0.2718	OOM	0.2405	OOM	0.2861	**0.4194**
NMI											
COIL20	0.8053	0.9407	0.8639	0.8130	0.6596	0.8331	0.9520	0.7018	0.8840	0.9384	**1.0000**
MSRC-v6	0.7757	0.8200	0.7756	0.7645	0.7809	0.8465	0.7617	0.8202	0.8653	0.8359	**0.9067**
3Sources	0.5581	0.6216	0.5557	0.5166	0.0830	0.5183	0.7856	0.6358	0.6980	0.6085	**0.8623**
Digit4k	0.6016	0.6944	0.3567	0.5754	0.6170	0.4229	0.6913	0.6892	0.5745	0.7970	**0.8106**
Caltech101-7	0.5147	0.6595	0.4920	0.5124	0.4176	0.5748	0.7400	0.5363	0.5306	0.6095	**0.7411**
Caltech101-all	0.1066	0.3446	0.4495	0.1569	0.3438	0.4479	OOM	0.2970	OOM	0.4881	**0.5555**
Purity											
COIL20	0.7542	0.7819	0.8063	0.7715	0.5542	0.7910	0.8910	0.5979	0.8403	0.8958	**1.0000**
MSRC-v6	0.8571	0.7668	0.8286	0.8238	0.8619	0.9190	0.8095	0.8429	0.9190	0.9143	**0.9571**
3Sources	0.7751	0.7460	0.7574	0.7278	0.3964	0.7811	0.8698	0.7633	0.8410	0.7692	**0.9408**
Digit4k	0.7200	0.5906	0.6163	0.5532	0.8255	0.6465	0.7450	0.8665	0.7785	0.9173	**0.9230**
Caltech101-7	0.8596	0.5943	0.7008	0.8670	0.6716	0.7151	0.8711	0.8236	0.8935	0.8915	**0.8948**
Caltech101-all	0.1371	0.1953	0.4427	0.2096	0.3389	0.3463	OOM	0.2969	OOM	**0.4773**	0.4631

Bold font indicates the best results and underlining indicates the second-best.

effects of randomness, all methods were executed 10 times, and average results were reported. We selected three metrics to evaluate clustering performance: accuracy (ACC), purity (PUR) and normalized mutual information (NMI).

D. Clustering Performance

As shown in Table III, Our proposed PBDMSC method achieves state-of-the-art clustering performance on six real-world datasets, significantly outperforming existing subspace-based and deep neural network based multi-view clustering algorithms. On the COIL20 dataset, PBDMSC attains perfect clustering accuracy. It also excels on other datasets like MSRC-v6, 3Sources, and Caltech101-7, demonstrating its effectiveness in learning discriminative multi-view representations. The results highlight the advantages of PBDMSC's key components, including pseudo-label guided contrastive learning and bidirectional attention-based self-representation learning, in enhancing multi-view clustering performance.

E. Running Time Analysis

To provide a comprehensive evaluation of computational efficiency, we compare the running time of PBDMSC with several representative multi-view clustering methods, including four deep learning-based approaches (MCGC [24], MFLVC [25], DMVSC [22], GCCL [16]) as well as the bidirectional probabilistic subspace multi-view clustering method (BPSA) [35]. All experiments were conducted using identical hardware configurations, with all deep learning methods trained for the same number of epochs. The running time comparison across six datasets is shown in Fig. 3. While PBDMSC exhibits a theoretical complexity, its practical runtime remains competitive with existing methods. These results demonstrate that our method not only achieves competitive runtime performance but also preserves its core innovation in representation learning, making it feasible for real-world applications.

F. Ablation Study

In this section, we designed three variants of PBDMSC for ablation studies:

- $\text{PBDMSC}_{\text{w/o BA}}$: remove the bidirectional attention module, where the model learns the self-representation matrix solely based on low-dimensional embeddings, disregarding information from the affinity matrices of different views;
- $\text{PBDMSC}_{\text{w/o CL}}$: remove the contrast loss term L_c from (13) to analyze the impact of eliminating contrastive learning.
- $\text{PBDMSC}_{\text{w/o HEC}}$: remove the homogeneous expression constraints from (11).

As shown in Table IV, the removal of various components leads to varying degrees of performance degradation. Notably, the most significant drop in performance occurs when the bidirectional attention mechanism is removed, highlighting the importance of information from the affinity matrices of different views for model performance, thereby validating the effectiveness of our proposed bidirectional attention module. Additionally, the version without the contrastive learning loss term also experiences a substantial decline in performance, as the absence of this term prevents pseudo-labels from guiding the learning of low-dimensional feature embeddings, thereby limiting the acquisition of better low-dimensional embeddings. Lastly, the removal of the homogeneous expression constraint also results in decreased model performance, indicating its validity.

G. Parameter Analysis

Our algorithm primarily involves four hyperparameters. Specifically, λ_1 and λ_2 are used to balance the contrastive

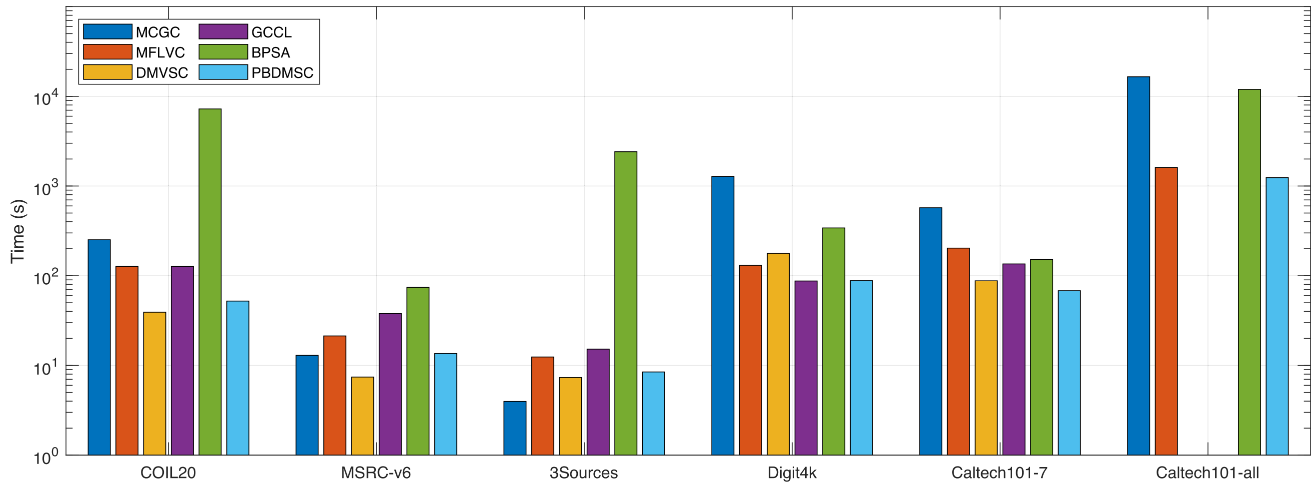


Fig. 3. Running time comparison (log scale) across six datasets. Missing bars indicate out-of-memory failures.

TABLE IV
ABLATION EXPERIMENT

Dataset	Metric	$PBDMSC_{w/o\ BA}$	$PBDMSC_{w/o\ CL}$	$PBDMSC_{w/o\ HEC}$	PBDMSC
COIL20	ACC	0.8056	0.9375	0.9806	**1.0000**
	NMI	0.8679	0.9693	0.9823	**1.0000**
	Purity	0.8257	0.9424	0.9806	**1.0000**
MSRC-v6	ACC	0.8381	0.9048	0.9524	**0.9571**
	NMI	0.7989	0.8419	0.9039	**0.9067**
	Purity	0.8381	0.9048	0.9524	**0.9571**
3Sources	ACC	0.7041	0.8817	0.9408	**0.9408**
	NMI	0.6308	0.7632	0.8607	**0.8623**
	Purity	0.7870	0.8817	0.9408	**0.9408**
Digit4k	ACC	0.8383	0.9093	0.9125	**0.9230**
	NMI	0.6940	0.7792	0.7896	**0.8106**
	Purity	0.8383	0.9093	0.9125	**0.9230**
Caltech101-7	ACC	0.7734	0.8412	0.8664	**0.8711**
	NMI	0.4281	0.5107	0.7113	**0.7411**
	Purity	0.8100	0.8440	**0.8989**	0.8948
Caltech101-all	ACC	0.1542	0.2206	0.3076	**0.4194**
	NMI	0.1951	0.4365	0.4476	**0.5555**
	Purity	0.2458	0.3453	0.3664	**0.4631**

The best results are indicated in bold font.

learning loss and self-representation learning loss, η balances intra-view and cross-view positive similarities in contrastive learning. the influence of positive and negative samples in contrastive learning, and γ balances the impact of the affinity matrices from various views in self-representation learning. In this section, we fix other parameters and discuss the effects of two sets of parameters on model performance. The values for λ_1 and λ_2 range from $\{0.001, 0.01, 0.1, 1, 10, 100\}$, η ranges from $\{0.1, 1, 100\}$, and γ ranges from $\{0.1, 1, 100, 1000\}$. We illustrate the performance impact of λ_1 and λ_2 across three datasets in Fig. 4, where it can be observed that the model is insensitive to the settings of λ_1 and λ_2.

We present the performance impact of η and γ across three datasets in Fig. 5, where a decrease in model performance is noted when γ is small. A smaller γ signifies reduced attention to the influence of the affinity matrices from different views, while a larger γ indicates that the model considers the affinity matrices from all views when learning the self-representation matrix. This reflects the effectiveness of our proposed bidirectional attention mechanism.

H. Visualization

1) Visualization of Subspace Self-Representation Matrices: To analyze the model's ability to learn discriminative self-representation matrices, we plotted the affinity matrices learned by MLRSSC, FPMVS_CAG and our model in Fig. 6 (the affinity matrix is derived from the self-representation matrix:$\mathbf{A} = \frac{|\mathbf{C}|+|\mathbf{C}^T|}{2}$). For the MSRC-v6 dataset, the visualization results from the MLRSSC method show less distinct similarity and exhibit a wider range of noise. The FPMVS_CAG method shows a less clear block diagonal structure, with significant interference from samples outside of the main categories. In contrast, our visualization clearly demonstrates a block diagonal structure

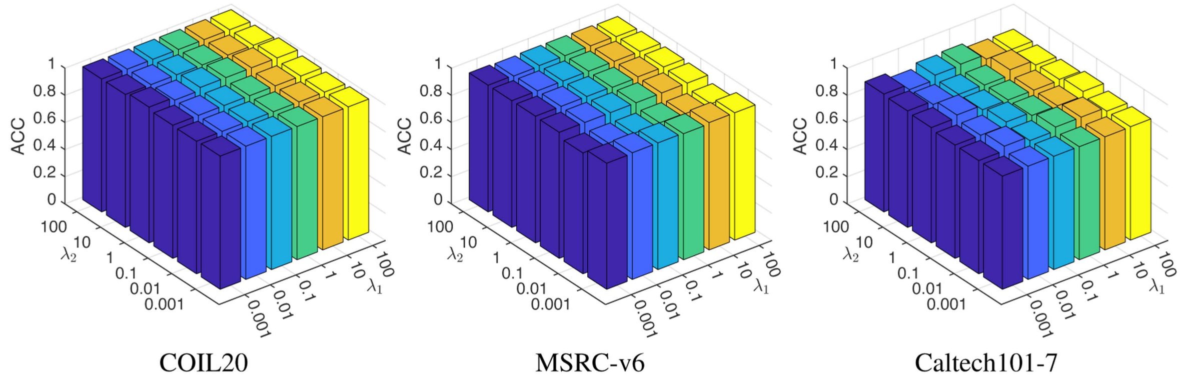


Fig. 4. The clustering ACC of PBDMSC under different λ_1 and λ_2.

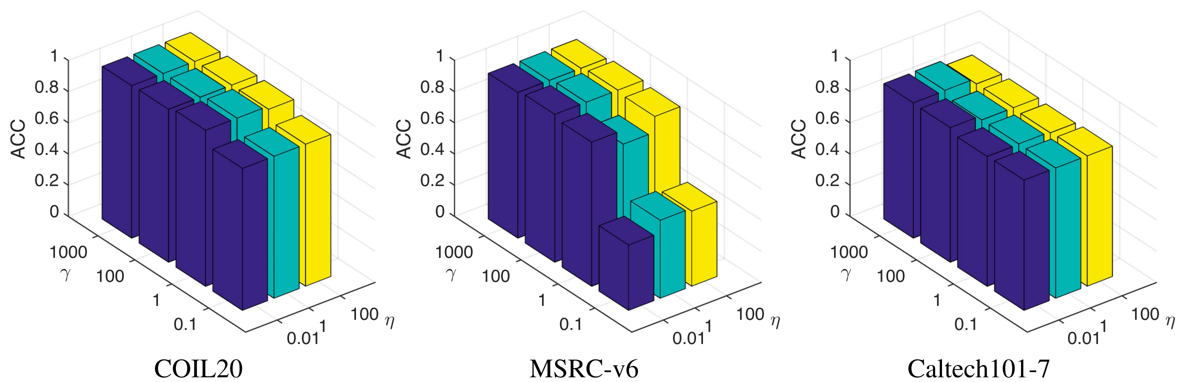


Fig. 5. The clustering ACC of PBDMSC under different η and γ.

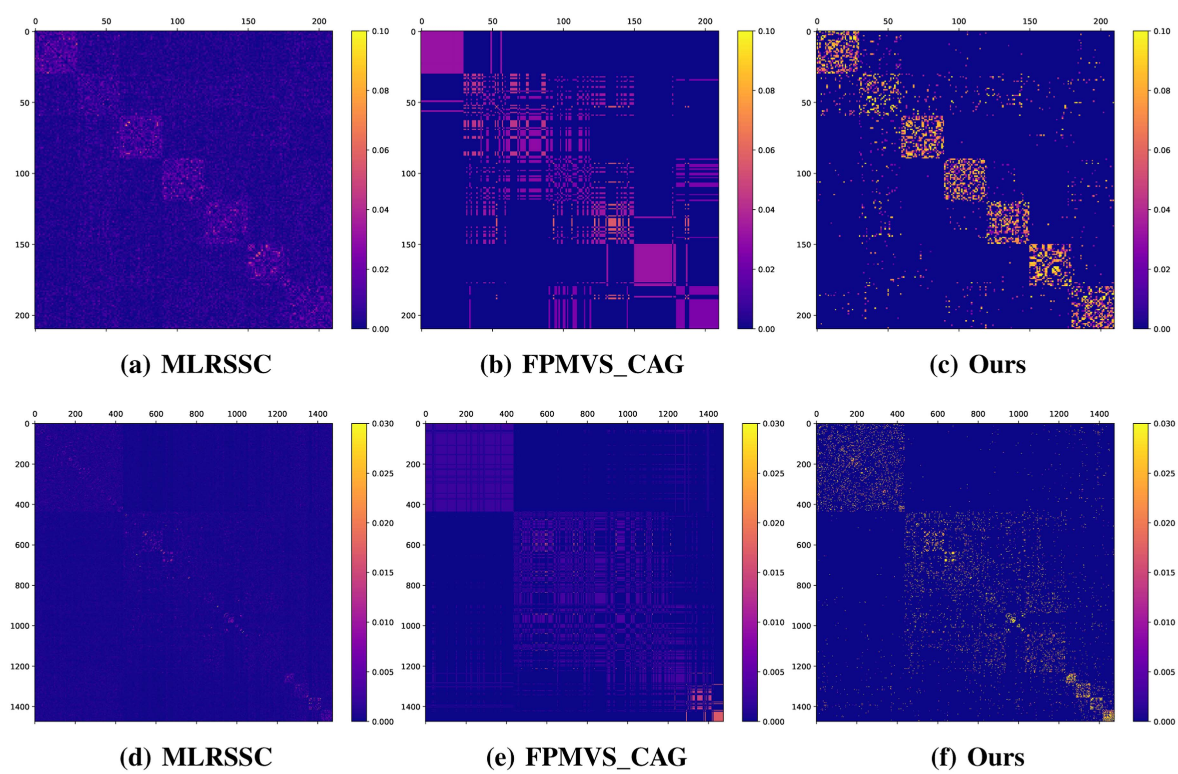
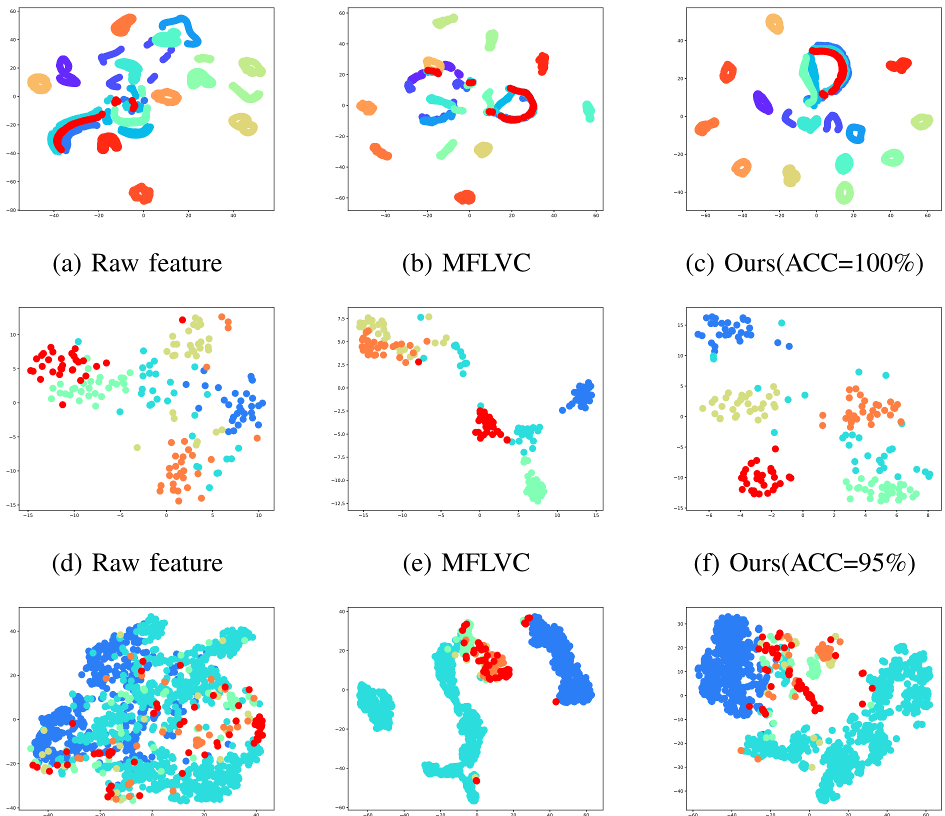


Fig. 6. The visualization of the affinity matrices between MLRSSC, FPMVS_CAG and ours on dataset MSRC-v6 (a)–(c) and Caltech101-7 (d)–(f).



(a) Raw feature (b) MFLVC (c) Ours(ACC=100%)

(d) Raw feature (e) MFLVC (f) Ours(ACC=95%)

(g) Raw feature (h) MFLVC (i) Ours(ACC=87%)

Fig. 7. The t-SNE visualization comparison between PBDMSC and MFLVC on the COIL20 (a)–(c), MSRC-v6 (d)–(f) and Caltech101-7 (g)–(i) datasets.

with seven small blocks, which corresponds precisely to the number of categories in the dataset. A similar pattern is observed for the Caltech101-7 dataset. The MLRSSC method struggles to reveal category relationships between different samples, while FPMVS_CAG only shows a rather coarse block diagonal structure. In comparison, our method produces a clear and distinct structure, where seven small blocks representing different categories can be seen in the plot. These results demonstrate that our proposed method is capable of learning a clearer and more structured affinity matrix, resulting in improved clustering performance. This reflects the model's ability to learn more discriminative self-representation matrices.

2) Embedding Visualization: To analyze the model's capability in learning feature representations, we performed t-SNE visualization on the original features of the COIL20, MSRC-v6 and Caltech101-7 datasets, as well as the features learned through MFLVC and our model. As shown in Fig. 7, the features learned through our network exhibit significantly better discriminative ability. On the COIL20 dataset, the features learned by our method almost completely separate all categories of samples, while the original features and those learned by the MFLVC method show much more overlap. For the MSRC-v6 dataset, the features learned by our method demonstrate greater dispersion between samples of different categories. On the Caltech101-7 dataset, similar results are observed, with our method showing overall better separability. The final clustering results are consistent with the effectiveness of feature representation learning, resulting in accuracy values of 100% and 95.24% on the COIL20 and MSRC-v6 datasets and 87.1% on Caltech101-7, respectively.This provides compelling evidence for the effectiveness of our proposed model, while promoting better feature representation learning through pseudo-label guided contrastive learning.

V. Conclusion

In this work, we have proposed a novel pseudo-supervised deep subspace clustering network PBDMSC to effectively learn latent representations and sample relationships from multi-view data. By integrating multi-view encoder, pseudo-label guided contrastive learning and discriminative self-representation learning modules, PBDMSC is able to capture the structured information from multiple views and enhance clustering performance. The key innovations include using a unified self-representation matrix to learn consistent relationships across views while aligning with the affinity matrix of each view, and employing pseudo-label guided contrastive learning to refine the learned embeddings. Furthermore, constraining the self-representation matrix using pseudo-labels helps avoid negative impacts from heterogeneous samples. In the future, we plan to further explore the potential of pseudo-labels in multi-view learning and extend PBDMSC to other applications such as multi-view classification and retrieval.

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